PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GORVERNMENT

LONGIDO DISTRICT COUNCIL

NAMANGA SECONDARY SCHOOL

FORM THREE MID TERM EXAMINATION.

CHEMISTRY 1

INSTRUCTIONS 23rd AUGUST 2023 TIME: 3:00 HRS

i. This paper consists of section A, B and C with a total of thirteen (13) questions.

ii. Answer **ALL** in section A and B and only two questions from section C.

iii. The following constants may be used

Atomic masses:H=1, C=12, N=14, O=16, Na=23, Cu=64, Zn= 65, Pb= 207, Ag=108, Cl=35.5

Avogadro's number=6.02×10^{23}

1Faraday= 96500 Coulombs

G.M. V=$22.4dm^3$

SECTION A(16marks)

1. For each of items (i)-(x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheets provided.

i. What is the maximum number of electrons in the innermost shell of an atom?

A. 3 B. 1 C. 4 D. 2 E. 8

ii. What feature is essential for a good fuel?

A. High speed of continuous energy supply
B. High energy value supplied
C. Low carbon dioxide supplied
D. High content of non-combustible material
E. Very high ignition point

iii. What conclusion can be drawn from the random movement of pollen grains suspended in air?

A. Matter is lighter in nature B. Matter is solid in nature C. Matter is particulate in nature
D. Matter is wave in nature E. Matter has no particles

iv. Which of the following is not a component of the First Aid Kit?

A. Goggles B. A pair of scissors C. Knife D. Gloves E. Razor blade

v. Which element is oxidized in the following reaction?

$2FeSO4 + Cl2 + H2SO4 \longrightarrow Fe(SO4)3 + 2HCl.$

A. Chlorine B. Hydrogen C. Oxygen D. Sulphur E. Iron

vi. Which of the following are the components needed to start fire?

A. Match box, fire wood and kerosene

B. Match box, fire wood and oxygen
C. Oxygen, fuel and fire wood
D. Oxygen, heat and match box
E. Oxygen, fuel and heat

vii. Why is nitrogen formed first during the fractional distillation of air?
A. It has got high boiling point
B. It has got low density
C. It has got low melting point
D. It has got high mass and density
E. It has got low boiling point

viii. A simple proof that chemical reactions is taking place in our bodies is that,
A. We eat a balanced diet
B. Doctors tell us in the hospital
C. We occasionally fall sick
D. The food we eat and liquids we drink differ from waste products we excrete
E. We were created so by God

ix. The subatomic particles which are not found in the nucleus of an atom are,
A. Protons B. Electrons C. Neutrons D. Nucleons E. Protons and Neutrons

x. The solid compound Q was put on a watch glass and left in air. After 40 minutes the compound was found to have water but not form a solution. This compound is said to be;
A. Soluble
B. Deliquescent
C. Hydroscopic
D. Efflorescent
E. Water used was temporary hard

2. Match the meaning of some properties of water in list A with the relevant situation in list B by writing the letter of the correct response beside the item number in the answer sheets provided.

LIST A	LIST B
i. Water has high surface tension	A. It forms drop
ii. Water is universal solvent	B. Fish can be able to see in water
iii. Water expands when freezes	C. Water has no any effect on litmus paper
iv. Water has high specific index	D. A bottle full of water tend to burst when left in deep freezer over night
v. Water is colorless	E. Water absorbs heat before it boils
vi. Pure water is neutral	F. Water can dissolve salts, sugar, acids and bases

SECTION B (54marks)

Answer **ALL** questions in this section, each question carries **9marks**

3. a. You are provided with a solution in bottle labelled A, B, C and D attached with P^H of

5.3, 7.3, 13 and 7. What will be the effect on the litmus paper when dipped into the given solutions?

Fill the table below

SOLUTION	EFFECT ON RED LITMUS PAPER	EFFECT ON BLUE LITMUS PAPER
A		
B		
C		
D		

b. i. What is acidity of a base?

ii. Provide the suitable indicator to be used in the titration of Ethanoic acid with ammonia solution.

4. a. What is volumetric analysis?

b. What volume of hydrochloric acid with 37.1% purity and density of 1.2g/cm^3 should be used to prepare 1litre of 0.1000M of HCl solution.

5. a. What is chemical equilibrium?

b. Read the table below and answer the questions that follows

experiment number	first reactant	second reactant	product	condition of the wall of the beaker containing mixtures	total number of moles of reactant(s)	total number of moles of product(s)	enthalpy change
1	A	B	M	Hot	4	2	46
2	C	D	N	Cold	1	4	290
3	E	L	P	Hot	2	2	76
4	F	G	R	cold	3	2	88

Questions

i. Write the balanced reversible reactions of each experiment by showing the enthalpy changes.

ii. identify experiments which are exothermic and which are endothermic.

iii. Identify the experiments whose position of equilibrium can be affected by pressure.

6. a. What is hardness of water? b. mention two merits of water hardness

c. Describe any three methods of softening hard water

7. a. Define the term chemical equation

b. Describe any two types of chemical reaction with example.

c. Complete and balance the following chemical reactions

i. $MCO_{3(S)} + HCl(aq) \longrightarrow$

ii. $MgCl + AgNO_3 \longrightarrow$

iii $CaO(s) + H_2O(l) \longrightarrow$

8. a. Define stoichiometry

b. How many moles of nitrogen gas would react with excess hydrogen gas to produce 2.24dm3 of ammonia gas at STP?

c. State Avogadro's law

SECTION C (30 marks)

Answer only two (2) questions from this section

9. a) i. Mention three factors affecting the selective discharge of ions on electrode during electrolysis

ii. Mention three factors affecting the magnitude of a substance liberated at electrode during electrolysis**.**

iii. State Faraday's first law of electrolysis.

b. Copper (II) sulphate solution undergo electrolysis by using carbon electrodes

i. write the chemical formula of the electrolyte used
ii. mention four important ions present in the solution
iii. show the reactions taking place at each electrode and identify the main product on cathode and anode.
iv. write the overall ionic equation for the electrolysis above
v. Draw the electrolytic cell for this electrolysis.

10. In an experiment of investigating the effect of concentration on the rate of chemical reaction between sodium thiosulphate and dilute hydrochloric acid by observing the disappearance of cross mark, the following results were obtained.

Volume of $Na_2S_2O_3$ (cm^3)	**Volume of water(cm^3)**	**Total volume(cm^3)**	**Time, t(s) taken for the cross to disappear**	**Rate of reaction, 1/t(s^-)**
30	0	30	0	
25	5	30	12	
20	10	30	36	
15	15	30	72	
10	20	30	152	
5	25	30	280	

a. Complete the table above
b. Write a balanced chemical equation of the experiment above
c. Write the ionic equation of the reaction above.
d. Plot a graph of volume of $Na_2S_2O_3$ against rate of reaction.

11. a. What is extraction of metals?
b. With the aid of clear labelled diagram and equations explain how sodium metal is extracted from its chief ore.